

Dipal Thaker

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PROFESSIONAL SUMMARY

Seeking SDE, AI/ML Engineer, or Forward Deployed Engineer roles. Software engineer who genuinely enjoys building things people actually use, from healthcare tools that help real patients to platforms that make other engineers faster. Comfortable in moving between frontend and backend, I pick up new tools quickly, and care more about whether something works reliably. Have spent the last two years learning by building, and I'm looking for a team where I can keep doing exactly that.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL

Frontend: React, TypeScript, Redux, Responsive Design

Backend: FastAPI, Spring Boot, REST APIs, Microservices

AI and LLM: Azure OpenAI, Groq LLaMA, LlamaIndex, RAG Pipelines, Evaluation Frameworks

Data: PostgreSQL, MySQL, Kafka, Data Pipelines, Query Optimization

Cloud and Infrastructure: AWS, Docker, GitHub Actions, CI/CD

Testing: Pytest, Jest, Unit Testing, Integration Testing, Code Review

EDUCATION

University of Illinois Urbana-Champaign | *MS in Information Management (STEM), GPA 3.83/4.00*

Aug 2024 - May 2026

K. J. Somaiya College of Engineering | *B.Tech in Information Technology, GPA 3.40/4.00*

Jun 2019 - Jul 2023

PROFESSIONAL EXPERIENCE

Business Intelligence Group, UIUC | *Applied AI Engineer*

Aug 2025 – Dec 2025

- Built and shipped 2 production AI systems for healthcare billing and denial workflows using Azure OpenAI and LlamaIndex, cutting hallucinated or out of scope responses by 95 percent through better retrieval design and tighter prompt constraints.
- Designed an evaluation framework in Python that tested 200 plus real interactions, catching unreliable outputs before they reached anyone relying on the system.
- Reduced projected support escalations by 30 percent by building fallback logic in the LLM pipeline that knew when to hand a problem to a human instead of guessing.
- Added PII redaction access controls using Python based validation rules so sensitive patient data stayed protected throughout the entire pipeline.
- Worked directly with healthcare stakeholders to turn vague requirements into something engineers could actually build, then explained the results back in plain language.

School of Information Sciences, UIUC | *Graduate Research Assistant, Full Stack Engineer*

Mar 2025 – May 2026

- Built a full stack health app from scratch using React, TypeScript, and Flutter for 100 plus elderly users, owning everything from the PostgreSQL database to the screens people actually tapped on.
- Increased user engagement by 35 percent by simplifying navigation and adding personalized recommendations built on usage data, not guesses.
- Built a real time data pipeline using Kafka and FastAPI that processed information in under 500 milliseconds, then improved PostgreSQL query performance by 40 percent once it started slowing down under load.
- Set up automated testing with Pytest and Jest, and deployment pipelines with GitHub Actions, so the team could ship changes daily without breaking what already worked.
- Led a 5 person team through sprint planning and code reviews, helping teammates write cleaner, more confident Python and TypeScript code.

Deloitte Consulting | *Software Developer*

Jun 2023 – Aug 2024

- Rebuilt a healthcare data pipeline using SQL and Python that was taking too long to run, cutting execution time in half through better query design and cleanup of redundant steps.
- Caught and fixed recurring data quality issues by building automated validation checks in SQL, reducing manual review work by 30 percent for the team downstream.
- Sat with clients to understand faults in their systems, shipped backend fixes in Java and Spring Boot that held up under real production load.
- Owned production incidents end to end, tracking down root causes across the application, database, deployment layers until actually resolved.

PROJECTS

Job Application Tracker | *React, TypeScript, FastAPI, PostgreSQL*

Oct 2025

- Built a web app where users log job applications, track status (applied, interviewing, offer, rejected), store notes and document links, and filter by company or stage, replacing a messy spreadsheet.
- Shipped it solo end to end including JWT login, 10 plus REST endpoints, and a PostgreSQL schema, cutting personal time spent organizing applications by 30 to 40 percent.

Mantis Agentic Platform | *Python, TypeScript, FastAPI, Azure OpenAI, PostgreSQL*

Jun 2025

- Built a multi step AI agent using Azure OpenAI and Python that takes a coding task, plans the steps needed, writes and tests code, and pauses to ask a human to approve risky changes before continuing, instead of failing silently or guessing.
- Used FastAPI and PostgreSQL to log every tool call and retrieval step the agent took, cutting task failure rate by 40 percent by turning vague failures into specific, fixable bugs.